figure-drafts

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# arranging plots
library(patchwork)

# coloring
library(wesanderson)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
```

```
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning and Wrangling

Cleaning

`sites` object

```
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

`taxon_list` object

```
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake")) |>
  select(verbatim_name, scientificName)
```

`water_quality` object

```
water_quality_clean <- water_quality |>
  clean_names() |>
  semi_join(ncos_sites, by = "site") |>
  unite("date_site", date, site, remove = TRUE) |>
  group_by(date_site) |>
  summarize(med_ph    = median(p_h, na.rm = TRUE),
```

```
            med_sal    = median(salinity_ppt, na.rm = TRUE),
            med_do     = median(dissolved_oxygen_mg_l, na.rm = TRUE),
            med_temp   = median(temperature_c, na.rm = TRUE)) |>   # <-- ADD THIS
  ungroup() |>
  filter(date_site != "2022-08-12_NVBR")
```

aq_ins object

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         cladocera) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:cladocera,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
```

```r
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full)) |>

  # ADD: log+1 transform of average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>                        # <-- Fig 1, 2, 3, 4

  # ADD: season derived from date
  mutate(season = case_when(
    month(date) %in% c(3, 4, 5)  ~ "Spring",
    month(date) %in% c(6, 7, 8)  ~ "Summer",
    month(date) %in% c(9, 10, 11) ~ "Fall",
    month(date) %in% c(12, 1, 2) ~ "Winter"
  ),
  season = factor(season, levels = c("Spring", "Summer", "Fall", "Winter"))) |># <-- Fig 4
  mutate(taxon = factor(taxon, levels = c("copepod", "ostracod", "cladocera")))
```

Figure 1

Figure 2 - Mean and Distribution of Invertebrate Abundance (log +1/L) for Three Tax

```r
jitter <- ggplot(data = aq_ins_clean,
                 aes(x = taxon,
                     y = log_ave_lit,
                     color = taxon)) +

  geom_jitter(height = 0,
              alpha = 1,
              width = 0.2,
              shape = 21) +
```

```
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "mean",
               size = 4,
               color = "black",
               shape = 18) + # 18 = filled diamond
  labs(x = "Taxon",
       y = "log + 1 transformed average abundance/L",
       title = "Mean and Distribution of Invertebrate Abundance (log +1/L) for Three Tax")

jitter
```

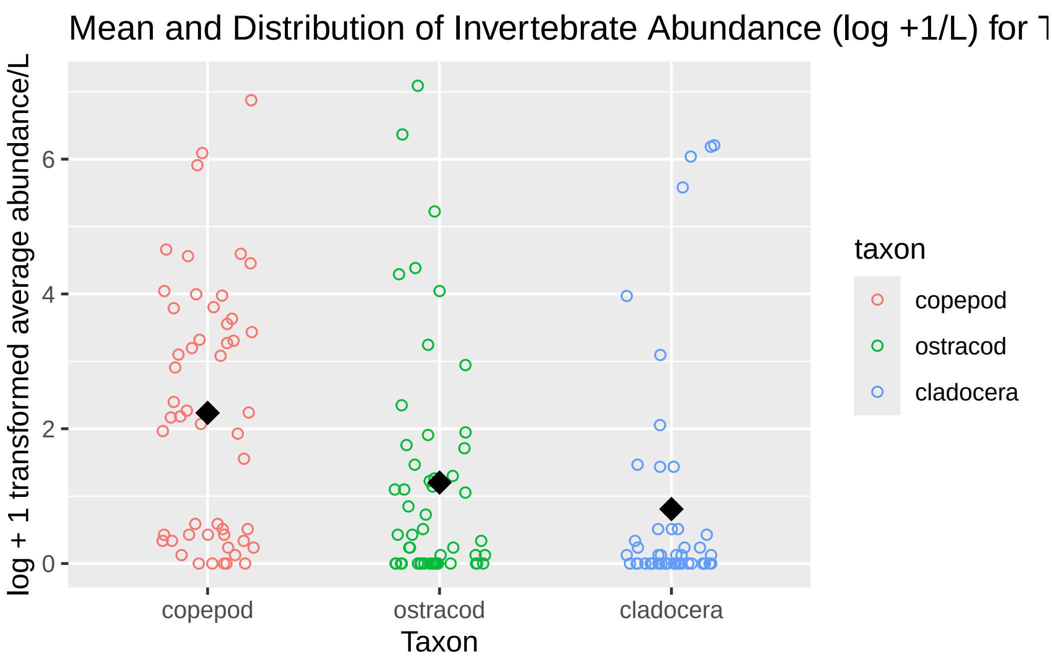

Figure 3

Figure 4